

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 3: Properties of Discrete-Time Systems

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Last Time

Systems building blocks.

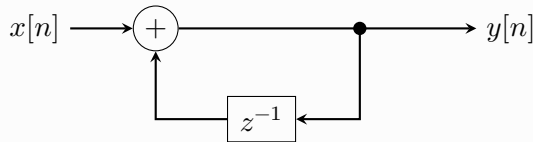
Example (FIR Filter).

A finite impulse response system, as introduced last lecture.

Example (Accumulator System).

$$y[n] = \sum_{\ell=-\infty}^n x[\ell], \quad y[n-1] = \sum_{\ell=-\infty}^{n-1} x[\ell] \implies y[n] - y[n-1] = x[n].$$

This can be calculated practically as $y[n] = y[n-1] + x[n]$ (given some initial condition):



The impulse response is $h[n] = H\{\delta[n]\} = u[n]$. This system is **not FIR**; it is **IIR** (Infinite Impulse Response).

Properties we will study: [Linearity](#), [Time Invariance](#), [Causality](#), [Impulse Response](#), [Convolution](#), [Stability](#).

2 1 Linear Discrete-Time Systems

Definition (Linearity).

For arbitrary $\alpha, \beta \in \mathbb{C}$ and arbitrary inputs $x_1[n]$ and $x_2[n]$: if

$$y_1[n] = H\{x_1[n]\} \quad \left(x_1[n] \xrightarrow{H} y_1[n] \right)$$

and if

$$x_2[n] \xrightarrow{H} y_2[n],$$

then if

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

we must have

$$x[n] \xrightarrow{H} y[n] = \alpha y_1[n] + \beta y_2[n].$$

Example (Accumulator is Linear).

$$y[n] = \sum_{\ell=-\infty}^n x[\ell] \quad (\text{Accumulator})$$

$$x_1[n] \xrightarrow{H} y_1[n] = \sum_{\ell=-\infty}^n x_1[\ell]$$

$$x_2[n] \xrightarrow{H} y_2[n] = \sum_{\ell=-\infty}^n x_2[\ell]$$

If

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

then

$$\begin{aligned} x[n] \xrightarrow{H} y[n] &= \sum_{\ell=-\infty}^n (\alpha x_1[\ell] + \beta x_2[\ell]) \\ &= \alpha \sum_{\ell=-\infty}^n x_1[\ell] + \beta \sum_{\ell=-\infty}^n x_2[\ell] \\ &= \alpha y_1[n] + \beta y_2[n]. \end{aligned}$$

Therefore, the accumulator is linear.

3 2 Homogeneity for the Accumulator, via the Difference-Equation Form

Example.

$$y[n] = y[n - 1] + x[n] \quad (\text{accumulator})$$

Assume

$$x[n] \xrightarrow{H} y[n]$$

Let

$$x_1[n] = \alpha x[n] \xrightarrow{H} y_1[n]$$

Then

$$y_1[n] = y_1[n - 1] + \alpha x[n], \quad \alpha y[n] = \alpha y[n - 1] + \alpha x[n].$$

Subtracting,

$$\begin{aligned} y_1[n] - \alpha y[n] &= y_1[n - 1] - \alpha y[n - 1] \\ \Rightarrow g[n] &= g[n - 1], \quad g[n] \triangleq y_1[n] - \alpha y[n]. \end{aligned}$$

Remark.

$g[n]$ should be zero for all n for homogeneity to hold.

We assume **zero initial conditions (ICs)** (a restrictive definition; we will use something more general for LTI systems):

$$\begin{aligned} \exists M \in \mathbb{Z} \text{ s.t. } y[n] &= 0, \quad x[n] = 0, \quad y_1[n] = 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n. \end{aligned}$$

So $y_1[n] = \alpha y[n]$ for all n : **Homogeneity** ✓.

Remark.

Proving linearity is frequently harder than disproving linearity.

Example (Median Filter: Homogeneous but not Additive).

$$y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}.$$

- Homogeneity? ✓
- Additivity? ✗

For single impulses $x_1[n]$ and $x_2[n]$ at different locations, the median filter outputs $y_1[n]$ and $y_2[n]$ that are each zero (or small); but for $x_1[n] + x_2[n]$ (two nonzero samples), the median output $y[n] \neq y_1[n] + y_2[n]$. Hence **additivity fails**.

5 3 Time-Invariant Discrete-Time Systems

Definition (Time-Invariance).

For arbitrary $x[n] \xrightarrow{H} y[n]$: if $x_1[n] = x[n - n_0]$ for arbitrary $n_0 \in \mathbb{Z}$, then $x_1[n] \xrightarrow{H} y_1[n] = y[n - n_0]$.

Example (FIR Filter is Time-Invariant).

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] \quad (\text{FIR filter}).$$

Let $x_1[n] = x[n - n_0]$. Then

$$y_1[n] = \sum_{k=0}^{M-1} \alpha_k x_1[n-k] = \sum_{k=0}^{M-1} \alpha_k x[(n-k)-n_0] \stackrel{?}{=} y[n-n_0] \implies \text{time invariant!}$$

Example (A System that is Not Time-Invariant).

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0].$$

For $x[n] = \delta[n]$:

$$y[n] = \begin{cases} \alpha_n + 1, & 0 \leq n \leq M-1 \\ 0 + 1, & \text{else} \end{cases}.$$

For $x_1[n] = \delta[n-1]$:

$$y_1[n] = \begin{cases} \alpha_{n-1} + 0, & 1 \leq n \leq M \\ 0 + 0, & \text{else} \end{cases}.$$

Since $y_1[n] \neq y[n-1]$ (the added constant $x[0]$ does not shift), **time invariant?**
Nope.

The extra term always reads the input at index 0, not at a lag relative to n . Shifting the input by n_0 therefore does not translate that contribution through time; the output is not a shifted copy of the original response.

4 Linear Time-Invariant (LTI) Systems

- Linear
- Time-invariant
- Easy to analyze, characterize, implement, and design.

4.1 Important Subclass: Linear Constant-Coefficient Difference Equations (LDEs)

An important subclass, with zero ICs:

$$\sum_{k=0}^{N-1} d_k y[n-k] = \sum_{q=0}^{M-1} p_q x[n-q].$$

Solving for $y[n]$:

$$y[n] = \sum_{k=1}^{N-1} \tilde{d}_k y[n-k] + \sum_{q=0}^{M-1} \tilde{p}_q x[n-q],$$

$$\left(\tilde{d}_k = -\frac{d_k}{d_0}, \quad \tilde{p}_q = \frac{p_q}{d_0}; \text{ assuming } d_0 \neq 0 \right).$$

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Examples: Accumulator (FIR Filter).

$$y[n] = y[n-1] - x[n-1] + \rho(y[n-1] - x[n-1]),$$

where $y[n]$ is the loan balance, $y[n-1]$ the previous balance, $x[n-1]$ the payment, and ρ the interest rate. Similar in spirit to the continuous-time ODE form

$$y(t) = \sum_{n=0}^N \alpha_n \frac{d^n x(t)}{dt^n} + \sum_{m=1}^M \beta_m \frac{d^m y(t)}{dt^m}.$$

5 Causal Systems

Definition (Causality).

The m th output sample $y[m]$ depends only on input samples $x[n]$ with $n \leq m$. This must be true for all possible inputs and all m .

Examples.

- $y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}$: **not causal**.
- $y[n] = y[n-1] + x[n]$: causal, with an IC.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k]$: **causal**.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0]$: **not causal**.

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Property (Causal Systems and Matching Inputs).

For a causal system, if $x_1[n] \xrightarrow{H} y_1[n]$ and $x_2[n] \xrightarrow{H} y_2[n]$, and if $x_1[n] = x_2[n]$ for all $n < N$, and the ICs are the same, then $y_1[n] = y_2[n]$ for all $n < N$.

6 Stability

Definition (BIBO Stability).

A system is **bounded-input bounded-output (BIBO) stable** if every bounded input yields a bounded output, i.e., if $|x[n]| \leq B_x < \infty$ for all n , then $|y[n]| \leq B_y < \infty$ for all n .

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Example (Moving Average is BIBO Stable).

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k].$$

Assume

$$|x[n]| \leq B_x < \infty \quad \text{for all } n.$$

Then

$$\begin{aligned} |y[n]| &= \left| \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \right| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} B_x = B_x \implies \text{BIBO stable!} \end{aligned}$$

Example (Accumulator is Not BIBO Stable).

$y[n] = y[n-1] + x[n]$, with IC $y[n] = x[n] = 0$ for $n < 0$. Let $x[n] = u[n]$ (bounded). Then $y[n] = (n+1)u[n]$, which is **not bounded**.

7 Impulse Response

Definition (Impulse Response).

The impulse response $h[n]$ is the output of a discrete-time system to the input $\delta[n]$.

Example (Accumulator).

$$y[n] = \sum_{k=-\infty}^n x[k] \implies h[n] = \sum_{k=-\infty}^n \delta[k] = u[n].$$

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Fact.

LTI systems with zero ICs are **completely characterized** by their impulse responses.

Remark.

A more general definition of zero ICs for LTI systems: if $x[n] = 0$ for all n , then $y[n] = H\{x[n]\} = 0$ for all n : “zero-input” $\Rightarrow$ “zero-output.” (Non-zero ICs $\Rightarrow$ not LTI.)

Proof of fact: next time.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.5–4.7
- Monson H. Hayes, *Schaum’s Outline of Digital Signal Processing*, 2nd ed., Ch. 1.5, 2.4